Analysis

PENG

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1 Introduction

Definition 1.1 (analysis)**.** *Analysis of real numbers, sequences and series of real numbers, and real-valued functions [1, 2].*

2 Natural Number

Definition 2.1 (natural number)**.** *A natural number is any element of the set*

$$\mathbb{N} := 0, 1, 2, 3, \ldots \tag{2.1}$$

Axiom 2.1 (0)**.** *0 is a natural number.*

Axiom 2.2 (successor)**.** *If n is a natural number, then $n{+}{+}$ is also a natural number.*

Definition 2.2. $1 := 0{+}{+}$, $2 := 1{+}{+}$, $3 := 2{+}{+}$, ...

Axiom 2.3 ($n{+}{+} \neq 0$)**.** *0 is not the successor of any natural number n.*

Axiom 2.4 ($n{+}{+} \neq m{+}{+}$)**.** *Different natural numbers must have different successors.*
If $n \neq m$, then $n{+}{+} \neq m{+}{+}$. Equivalently A.1, if $n{+}{+} = m{+}{+}$, then $n = m$.

Axiom 2.5 (principle of mathematical induction)**.** *Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{+}{+})$ is also true. Then $P(n)$ is true for every natural number n.*

Lemma 2.1 (recursive definition of sequence)**.** *$a_0 = c$ and $a_{n{+}{+}=f_n(a_n)}$ for each natural number n.*

2.1 Addition

Definition 2.3 (addition)**.** *Define* $0 + m := m$*. Suppose inductively that we have defined* $n + m$*. Then* $(n{+}{+}) + m := (n + m){+}{+}$*.*

$a_n = n + m$ and $f_n(a_n) = a_n{+}{+}$

Lemma 2.2. $n + 0 = n$

Lemma 2.3. $n + (m{+}{+}) = (n + m){+}{+}$

Proposition 2.1 (addition is commutative)**.**

$$n + m = m + n \tag{2.2}$$

Proposition 2.2 (addition is associative)**.**

$$(a + b) + c = a + (b + c) \tag{2.3}$$

Proposition 2.3 (cancellation law)**.** *If* $a + b = a + c$*, then* $b = c$*.*

Definition 2.4. *A natural number* n *is positive if and only if* $n \neq 0$*.*

Lemma 2.4. *If* a *is a positive natural number and* b *is a natural number, then* $a + b$ *is positive.*

Lemma 2.5. *If* $a + b = 0$*, then* $a = 0$ *and* $b = 0$*.*

Lemma 2.6. *If* a *is a positive number, then there exists exactly one natural number* b *such that* $b{+}{+} = a$*.*

Definition 2.5 (order)**.**

- $n \geq m$ *or* $m \leq n$ *if and only if* $n = m + a$ *for some natural number* a
- $n > m$ *or* $m < n$ *if and only if* $n \geq m$ *and* $n \neq m$

Proposition 2.4 (properties of order)**.**

- *reflexive:* $a \geq a$
- *transitive: If* $a \geq b$ *and* $b \geq c$*, then* $a \geq c$
- *antisymmetric: If* $a \geq b$ *and* $b \geq a$*, then* $a = b$
- *addition preserve order:* $a \geq b$ *if and only if* $a + c \geq b + c$
- $a < b$ *if and only if* $a{+}{+} \leq b$
- $a < b$ *if and only if* $b = a + d$ *for some positive number* d

Proposition 2.5 (trichotomy of order)**.** *Exactly one of following statement is true:* $a < b$*,* $a = b$*, or* $a > b$*.*

Proposition 2.6 (strong principle of induction)**.** *Let* m_0 *be a natural number. Suppose that for each* $m \geq m_0$*, if* $P(m')$ *is true for all* $m_0 \leq m' < m$*, then* $P(m)$ *is also true. Then* $P(m)$ *is true for all* $m \geq m_0$*.*

Example 2.1 (Fibonacci sequence)**.**

$$F_0 = 0 \tag{2.4}$$

$$F_1 = 1 \tag{2.5}$$

$$F_n = F_{n-1} + F_{n-2} \tag{2.6}$$

2.2 Multiplication

Definition 2.6 (multiplication)**.** *Define* $0 \times m := 0$*. Suppose inductively that we have defined* $n \times m$*. Then* $(n++) \times m := (n \times m) + m$*.*

Lemma 2.7. $n \times 0 = 0 \times n$

Lemma 2.8. $n \times (m++) = (n \times m) + n$

Proposition 2.7 (multiplication is commutative)**.**

$$n \times m = m \times n \tag{2.7}$$

Lemma 2.9 (zero divisor)**.** $n \times m = 0$ *if and only if at least one of* n *and* m *is equal to 0. If* n *and* m *are both positive, then* nm *is also positive.*

Proposition 2.8 (distributive law)**.**

$$a(b+c) = (b+c)a = bc + ca \tag{2.8}$$

Proposition 2.9 (multiplication is associative)**.**

$$(ab)c = a(bc) \tag{2.9}$$

Lemma 2.10 (multiplication preserve order)**.** *If* $a < b$ *and* c *is positive, then* $ac < nc$*.*

Proposition 2.10 (cancellation law)**.** *If* $ac = bc$ *and* c *is positive, then* $a = b$*.*

Proposition 2.11 (Euclid's division lemma)**.** *Let* n *be a natural number and* q *be a positive natural number. Then there exist natural numbers* m *and* r*, such that* $0 \leq r < q$ *and* $n = mq + r$*.*

Proof. $0 = 0 \times q + 0$ and $0 \leq r < q$. Suppose that $n = mq + r$ and $0 \leq r < q$. Then $n++ = mq + (r++)$ and $0 \leq r++ \leq q$. If $r++ = q$, $n++ = mq + q = (m+1)q + 0$. If $r++ < q$, $n++ = mq + r++$. □

Definition 2.7 (Exponentiation)**.** *Define* $m^0 := 1$*, i.e.* $0^0 := 1$*. Suppose recursively that* m^n *has been defined for some natural number* n*, then define* $m^{n++} := m^n \times m$*.*

3 Set Theory

Definition 3.1. *Set* A *is any unordered collection of objects.* $x \in A$ *if* x *lies in the collection, otherwise* $x \notin A$*.*

Axiom 3.1.

1. *If* A *is a set, then* A *is also an object.*
2. $A = B$ *if and only if every element of* A *belongs also to* B *and vice versa.*
3. *There exists an empty set* $\emptyset$*, such that every object* $x \notin \emptyset$*.*
4. *Let* A *be a non-empty set. Then there exist an object* $x \in A$*.*
5. *Singleton set* $\{a\}$*:* $y \in \{a\}$ *if and only if* $y = a$*. Pair set* $\{a, b\}$*:* $y \in \{a, b\}$ *if and only if* $y = a$ *or* $y = b$*.*
6. $x \in A \cup B \Leftrightarrow x \in A$ *or* $x \in B$*.*
7. $y \in \{x \in A : P(x)$ *is true*$\} \Leftrightarrow y \in A$ *and* $P(x)$ *is true.*
8. $z \in \{y : P(x, y)$ *is true for some* $x \in A\} \Leftrightarrow P(x, z)$ *is true for some* $x \in A$*.*
9. *There exists a set* $\mathbb{N}$*, whose elements are called natural numbers, and an object* $n++$ *assigned to every* $n \in \mathbb{N}$*, such that the Peano axioms* **??** *hold.*

Example 3.1. $\{a, a\} = \{a\}$

Example 3.2. $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$, $\{\emptyset, \{\emptyset\}\}$ *are all distinct.*

Proof. $y \notin \emptyset$.
$y = \emptyset \in \{\emptyset\}$.
$y = \{\emptyset\} \in \{\{\emptyset\}\}$.
$y = \emptyset$ or $y = \{\emptyset\} \in \{\emptyset, \{\emptyset\}\}$. □

Lemma 3.1. *Let a, b are objects, and A, B, C are sets.*

1. $\{a, b\} = \{a\} \cup \{b\}$
2. $A \cup B = B \cup A$
3. $(A \cup B) \cup C = A \cup (B \cup C)$

Definition 3.2.

$$\{a, b, c\} := \{a\} \cup \{b\} \cup \{c\}$$
$$\{a, b, c, d\} := \{a\} \cup \{b\} \cup \{c\} \cup \{d\}$$
$$\vdots$$

Definition 3.3 (Subsets)**.** 1. $A \subseteq B \Leftrightarrow$ *for any object* $x \in A$, *then* $x \in B$.

2. $A \subsetneq B$ *if* $A \subseteq B$ *and* $A \neq B$.

Lemma 3.2. *Let A, B, C be sets.*

1. *If* $A \subseteq B$ *and* $B \subseteq C$, *then* $A \subseteq C$.
2. *If* $A \subseteq B$ *and* $B \subseteq A$, *then* $A = C$.
3. *If* $A \subsetneq B$ *and* $B \subsetneq C$, *then* $A \subsetneq C$.

Definition 3.4. *Let A, B be sets.*

1. $A \cap B := \{x \in A : x \in B\}$.
2. $A \backslash B := \{x \in A : x \notin B\}$.

Lemma 3.3. *Let A, B, C be sets, and $\{A, B, C\} \subseteq X$.*

1. $A \cup \emptyset = A$ *and* $A \cap \emptyset = \emptyset$.
2. $A \cup X = X$ *and* $A \cap X = A$.
3. $A \cup A = A$ *and* $A \cap A = A$.
4. $A \cup B = B \cup A$ *and* $A \cap B = B \cap A$.
5. $(A \cup B) \cup C = A \cup (B \cup C)$ *and* $(A \cap B) \cap C = A \cap (B \cap C)$.
6. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *and* $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
7. $A \cup (X \backslash A) = X$ *and* $A \cap (X \backslash A) = \emptyset$.
8. *De Morgan laws:* $X \backslash (A \cup B) = (x \backslash A) \cap (X \backslash B)$ *and* $X \backslash (A \cap B) = (X \backslash A) \cup (X \backslash B)$.

Proof. Prop. 3.3 (5).
$x \in A \cap X \Leftrightarrow x \in A \Rightarrow A \cap X = A$. □

Proof. Prop. 3.3 (6).
$x \in A \cap (B \cup C) \Rightarrow x \in A$ and $(x \in B$ or $x \in C) \Rightarrow x \in A \cap B$ or $x \in A \cap C \Rightarrow x \in (A \cap B) \cup (A \cap C)$.
$x \in (A \cap B) \cup (A \cap C) \Rightarrow x \in (A \cap B)$ or $x \in (A \cap C) \Rightarrow x \in A$ and $x \in B \cup C \Rightarrow x \in A \cap (B \cup C)$. □

Lemma 3.4 (De mongan laws)**.**
!(A or B) = !A and !B.
!(A and B) = !A or !B.

Proof. Prop. 3.3 (8).
$x \in X \backslash (A \cup B) \Rightarrow x \in X$ and ($x \notin A$ and $x \notin B$).
$x \in X \backslash (A \cap B) \Rightarrow x \in X$ and ($x \notin A$ or $x \notin B$). □

Example 3.3.
$\{y : y = f(x)$ *for some* $x \in A\} = \{f(x) : x \in A\}$.
$\{x \in A : P(x)$ *is true*$\} \Rightarrow \{f(x) : x \in A; P(x)$ *is true*$\}$.

Example 3.4. $\{n + 3 : n \in \mathbb{N}, 0 \leq n \leq 5\} = \{8 - m : m \in \mathbb{N}, 0 \leq m \leq 3\}$.

Example 3.5 (Axiom 3.1 (8) ⇒ Axiom 3.1 (7))**.**
$y \in \{x : P(x)$ *is true for some* $x \in A\} \Rightarrow y \in \{x \in A : P(x)$ *is true*$\}$.

3.1 Functions

4 Power Series

A Tools

Definition A.1 (implication)**.** *"If P, then Q", then it is also true that "If not Q, then not P".*

References

[1] Terence Tao. *Analysis I.* Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.

[2] Terence Tao. *Analysis II.* Vol. 2. Springer, 2022. ISBN: 978-981-19-7284-3.